



Modeling

Entities and attributes



ER Logical Modeling

- Entity-Relationship Modeling

Entities are the equivalent of grammatical nouns, such as employees, departments, products, or financial accounts. An **entity** can be defined by means of its properties, called **attributes**.

Relationships are the equivalent of verbs or associations, such as the act of purchasing, the act of repairing, being a member of a group, or being a supervisor of a department.

- **Entities**

- are business concepts, and **not** tables.
- are made of **attributes**
- are related using **reference relationships**

Entities

- Composed of **attributes**
 - Either **simple** (built-in, LOV, user-defined type) or **complex** (complex type)
 - holds data quality rules (mandatory, LOV value range, etc.)
- **Entity type** defines how the entity matches records
 - **Fuzzy matched**: matches records from multiple sources based on data similarity.
 - **ID matched**: matches record from multiple sources using a common ID.
 - **Basic**: no matching, all records come from a single source or are authored in the data hub.
- The **primary key**
 - The attribute that identifies each **golden record**
 - Can be handled by the platform (UUID, Sequence, calculated using SemQL)
 - ... or loaded from the source systems
- **Reference relationships** relate entities

Entity Type

	 Basic	 ID-Matched	 Fuzzy-matched
Publishers	One or none	Multiple	Multiple
Match and Merge	Not supported Dups detected at creation time	Yes , using exact match on a shared ID Dups detected at creation time	Yes , using match rules Dups detected at creation time
Survivorship	Not supported	Consolidation is supported Value override is supported	Consolidation is supported Value override is supported
ID Management	Golden ID=Source ID	All source systems use a shared ID Golden ID=Source ID	Source systems have different IDs The Golden ID is generated
Recommended for	Centralized pattern Single source system Reference Data	Consolidation pattern, for entities with a shared ID.	Consolidation pattern, for entities with no shared ID.
Example	Product Family, Cost Centers	Products (with a shared SKU)	Customers, Contacts, Products (with no shared SKU)

Primary key attribute

- Primary Key (ID) identifies the master and golden records
- Behavior differs for the entity type
- **Basic and ID-Matched** entities
 - Golden and Master ID are the same
 - Generated by xDM (Sequence, UUID, Computed from other attributes)
 - or Manually entered/provided by source systems
- **Fuzzy-Matched** entities
 - Master and Golden records have a different IDs.
 - Golden ID
 - Generated by xDM (Sequence or UUID)
 - Master ID
 - Generated by xDM (Sequence, UUID, Computed from other attributes)
 - or Manually entered/provided by source systems



Names and Labels in Entities and Attributes

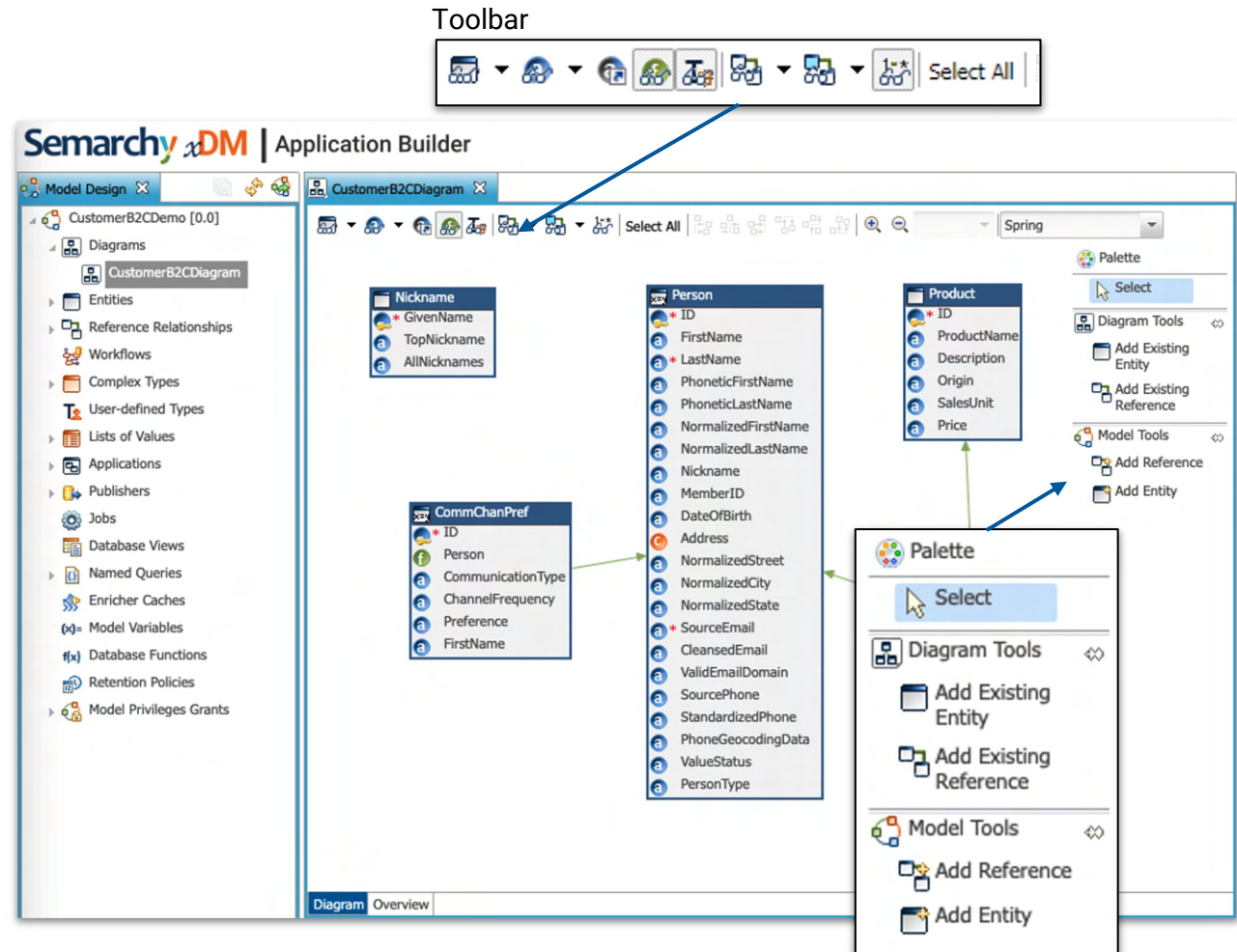
- Define simple and explicit **names** for entities and attributes
 - The name is used in SemQL rules and in the REST APIs.
 - E.g.: Customer, Address, BankDetail, etc.
- Define good **labels** and **plural labels** for entities and attributes
 - The labels appear in applications. This avoids having to override them in the applications.
- Add a **description** and a **documentation**
 - Descriptions are for the model designers, documentations appear to end-users.
- Define **icons** for entities
 - They are used to graphically represent the entity in the application.



Using good names and labels for clearly defined business concepts always save you time later.

Using diagrams

- Use **diagrams** to edit ER Models
 - Use multiple diagrams to view subsets of complex models
 - You can expose diagrams to users in the application documentation.
- Diagram editor
 - **Toolbar** for display options
 - Show entities and attributes names, labels
 - Show references cardinality, names, labels
 - **Palette**
 - Create new entities/references
 - Add existing entities to the diagram



Demo

Entity creation



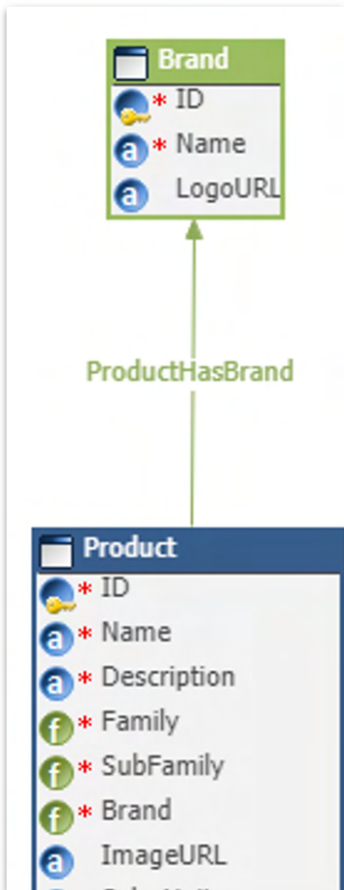
Modeling

Reference relationships



References Relate Two Entities

- Define a meaningful **Reference Name** and **Label**
 - e.g. *Product Has Brand*



Reference Relationship: ProductHasBrand

ProductRetailDemo ▾ ProductHasBrand ▾

Details

Brand

Name and Description

Auto Fill:	☒
Name:	ProductHasBrand
Physical Name:	PRODUCT_HAS_BRAND
Label:	Product Has Brand
Documentation:	
Description:	

Reference Relationships



- Reference relates two entities who have different **role** in relationship
 - **Referencing** Entity is the child entity - *Product*
 - **Referenced** Entity is the parent entity - *Brand*
- Define meaningful **Role Names**:
 - **Referencing Role Name** - *Products*
 - **Referenced Role Name** - *Brand*

The screenshot shows the configuration for the **ProductHasBrand** reference relationship in the Semarchy Intelligent Data Hub. The interface includes a breadcrumb trail: **ProductRetailDemo** > **ProductHasBrand**. The **Details** tab is selected, showing the **Brand** entity as the referencing entity. The configuration is divided into two main sections: **Name and Description** and **Referencing Entity (From)**. The **Referencing Entity (From)** section is expanded, showing the following fields:

Field	Value
Referencing Entity:	Product
Referencing Role Name:	Products
Referencing Role Label:	Product
Referencing Role Plural Label:	Products
Referencing Role Documentation:	

The **Referenced Entity (To)** section is also expanded, showing the following fields:

Field	Value
Referenced Entity:	Brand
Referenced Role Name:	Brand
Foreign Attribute:	Brand
Referenced Role Label:	Brand
Referenced Role Documentation:	
Physical Name:	BRAND

Reference Relationships

Define meaningful error messages

The screenshot shows the configuration interface for a reference relationship named 'ProductHasBrand'. The interface is divided into a left sidebar and a main configuration area. The sidebar contains a 'Details' tab and a 'Brand' icon. The main area is titled 'Reference Relationship: ProductHasBrand' and shows the relationship between 'ProductRetailDemo' and 'ProductHasBrand'. The 'Name and Description' section is expanded, showing fields for 'Auto Fill', 'Name', 'Physical Name', 'Label', 'Documentation', 'Description', 'Mandatory (One To Many)', 'Error Message (Mandatory)', and 'Error Message (Broken Reference)'. The 'Mandatory (One To Many)' checkbox is checked. The 'Error Message (Mandatory)' field contains the text 'A Brand must be selected!'. The 'Error Message (Broken Reference)' field contains the text 'This product brand does not exist!'.

Reference Relationship: ProductHasBrand	
ProductRetailDemo	ProductHasBrand
Details	
Brand	
Name and Description	
Auto Fill:	☐
Name:	ProductHasBrand
Physical Name:	PRODUCT_HAS_BRAND
Label:	Product Has Brand
Documentation:	
Description:	
Mandatory (One To Many):	☒
Error Message (Mandatory):	A Brand must be selected!
Error Message (Broken Reference):	This product brand does not exist!

- For empty references:

Brand

A Brand must be selected!

- For invalid references:

Found data validation issues

1 blocking issue found



This product brand does not exist!
Brand

References - Common patterns

- 1-N parent-child

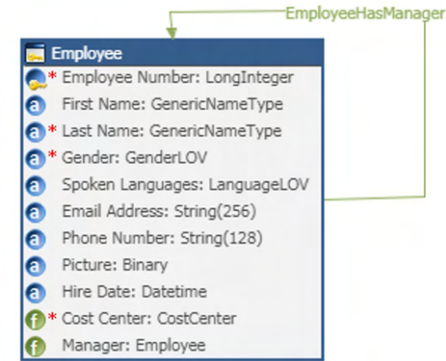


- Fixed-level hierarchy

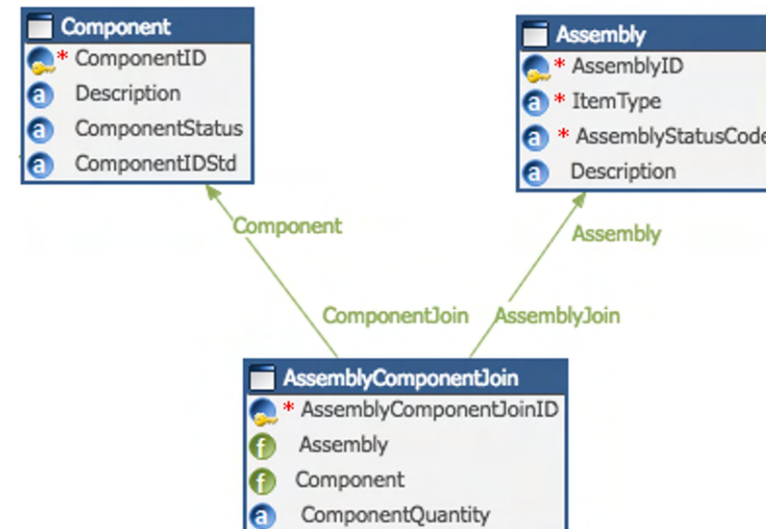


References - Common patterns

- Ragged hierarchy (self-join)



- Join table (N-M relationship)



Demo

References



Modeling

Record management



Record Management

Available actions and features for records depends on:

- The entity type (Basic, fuzzy or ID-matched)
- The reference relationships
- The rules applied to the entity and attributes
 - Enrichers, validations (E.g.: mandatory)
 - Matcher for fuzzy matched entities
 - Survivorship rules for ID and Fuzzy-matched entities
- The entity configuration
 - Historization
 - Delete

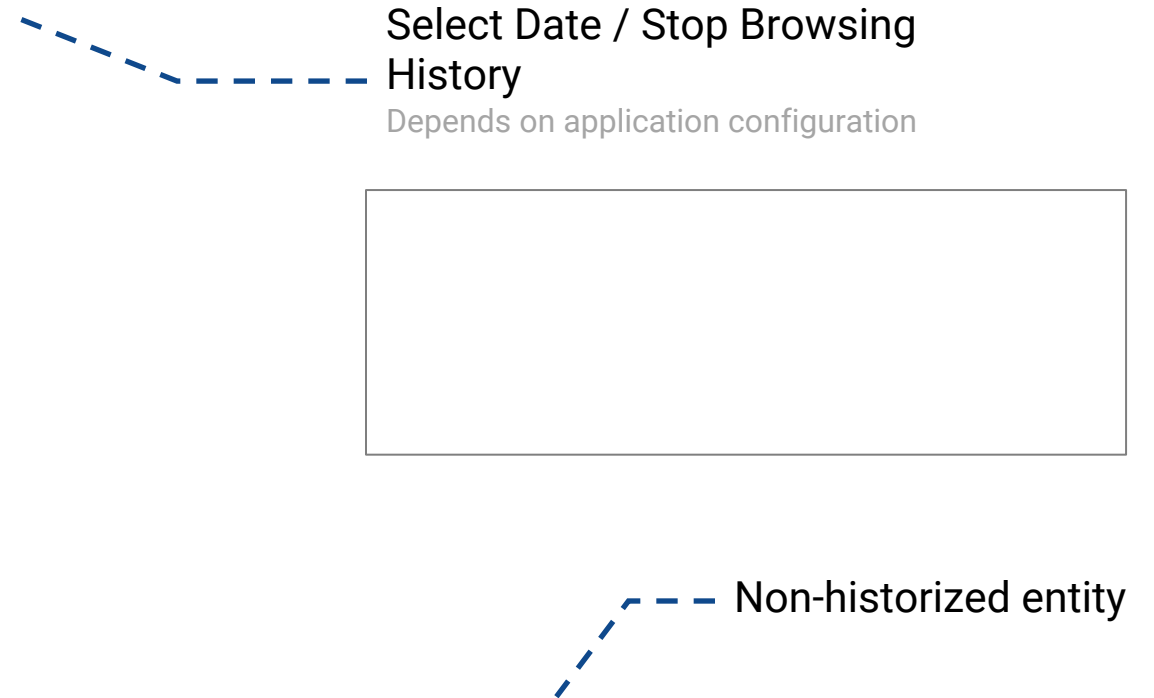
Historization

- Can be activated (optional) for each entity
 - Golden data can be historized for all entity types
 - Available for master record only with ID and fuzzy matched entities
- Allow users
 - to browse the history for a given record
 - browse the entire hub as of date.
- History is available in REST APIs and SQL APIs



Be careful with this feature as it may generate large volumes of data.

Browsing the history



Record to History Navigation

Enables querying and compound views, such as "Current record with history"